Math 534 H - Spring 2026 - Lecture 13

Continuing from (††) page 8 of Lecture 12.

we have integrating both sides w.r.t x

that

$$0 = \int_{-\infty}^{\infty} \Big(\underbrace{u_{tt} \cdot u_t}_{\frac{1}{2}\frac{d}{dt}(u_t)^2} \underbrace{- c^2 u_{xx} u_t}_{} + \underbrace{k u u_t}_{\frac{1}{2} k \frac{d}{dt}(u^2)} \Big) dx$$

Now we consider second term

$$\int_{-\infty}^{\infty} u_{xx} u_t \, dx \underset{\text{IBP}}{=} \underbrace{u_x \cdot u_t \Big|_{-\infty}^{\infty}}_{=0} - \int_{-\infty}^{\infty} \overbrace{u_x \cdot u_{xt}}^{\frac{1}{2}\frac{d}{dt}(u_x^2)} \, dx$$

as in bottom of pg 3 Lecture 12

$$c^2 = \frac{T}{\rho}$$

Then

$$0 = \frac{1}{2}\frac{d}{dt}\underbrace{\int_{-\infty}^{\infty} (u_t)^2 + c^2 (u_x)^2 + k u^2 \, dx}_{\tilde{E}[u](t)}$$

Then we get

$$\tilde{E}[u](t) = \tilde{E}[u](0)$$

$\hookrightarrow$ constant in time.

This energy is conserved in time.

Diffusion or Heat Equation in 1D $x \in \mathbb{R}$.

$$u_t - k u_{xx} = 0 \qquad k > 0 \qquad (H)$$

This is <u>NOT</u> time reversible.

Here is a bit harder to obtain a representation (and hence existence) of the solution.

Before we try to do so. Let's try to understand some properties of solutions (should these exist).

<u>Rmk</u>: Recall that IVP for (H) is

$$\begin{cases} u_t - k u_{xx} = 0 & x \in \mathbb{R}. \\ u(x,0) = \phi(x) \end{cases}$$

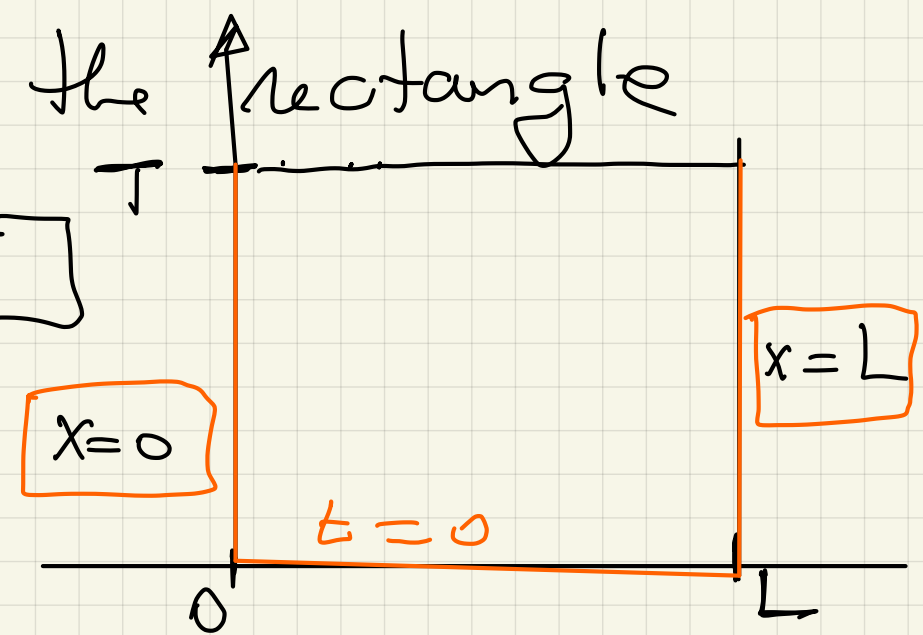
<u>MAXIMUM PRINCIPLE</u> : We consider

$$u_t - k u_{xx} = 0 \quad \text{for } x \in [0, L]$$

and let $t \in [0, T]$

So we consider (H) on the rectangle

$$R = [0, L] \times [0, T]$$



<u>Remark</u>: Uniqueness and stability of solution to IVP for (H) can be approached in two way: (1) using "energy method"; (2) Maximum principle.

There is NO maximum principle for the wave equation.

<u>THEOREM</u>: (weak <u>MAXIMUM PRINCIPLE</u>)

If $u(x,t)$ solves (H) in the closed rectangle $R = \underset{x}{[0,L]} \times \underset{t}{[0,T]}$. Then the maximum

value of $u(x,t)$ on $\mathcal{R}$ is achieved either initially $(t=0)$ or on the lateral sides $(x=0$ or $x=L)$. In other words

$$\max_{(x,t)\in\mathcal{R}}\{u(x,t)\} = \max_{(x,t)\in\sqcup}\{u(x,t)\}$$

$$\sqcup = \{(x,t) \,/\, t=0 \text{ or } x=0 \text{ or } x=L\}.$$

<u>Remark</u>: Once we prove the <u>max. ple</u> we'll also have proven the <u>minimum principle</u>

$$\min_{(x,t)\in\mathcal{R}}\{u(x,t)\} = \min_{(x,t)\in\sqcup}\{u(x,t)\}$$

because if $u(x,t)$ solves (H) then $-u(x,t)$ also solves (H) and

$$\min\{u(x,t)\} = -\max\{-u(x,t)\}$$

<u>Notation</u> Let's call the region $\sqcup = \Gamma$

<u>Proof</u>: A first natural (but ultimately unsuccesful) approach: u is continuous in (x,t) and R is closed so the $\max\{u(x,t)\}$ is indeed achieved on R

Suppose the maximum of $u(x,t)$ is achieved

at (x_0, t_0) inside R, then the gradient of u must vanished at that point: that is

$$u_t(x_0, t_0) = 0 = u_x(x_0, t_0)$$

If in addition we had that.

$$u_{xx}(x_0, t_0) < 0 \quad \underline{\text{strict inequality}}$$

this will give you a contradiction directly by plugging (x_0, t_0) in (H)

contradicting that u solves (H)

$$\underbrace{u_t(x_0, t_0)}_{=0} \underbrace{- k \underbrace{u_{xx}(x_0, t_0)}_{<0}}_{>0} > 0$$

$\boxed{\text{not } 0 \,!}$

What this tell us is that the maximum point (x_0, t_0) cannot be an internal point.

The problem is that "second derivative test" just gives you that $u_{xx}(x_0, t_0) \leq 0$